

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Fundamentals Of Data Science COURSE CODE: DSA04

COURSE OUTCOME COVERED

CO3: Evaluate datasets using descriptive statistics, including summarization, outlier detection, and distribution analysis, and perform statistical inference through estimation and hypothesis testing. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 20

Total Marks: 100

Annexure A

Data Interpretation

Code of Conduct:

I (Shaik Mohammed Abrar / 192472185) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Shaik Mohammed Abrar

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary) 5 Marks	Level 3 (Proficient) 3 /4 Marks	Level 2 (Developing) 2 Marks	Level 1 (Beginning) 1 Mark	Marks
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data	
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	

Annexure A

Problem 01: Hospital Patient Health Monitoring

Scenario

A hospital wants to investigate how patient lifestyle factors affect blood pressure levels.

Dataset

Patient_ID	Daily_Exercise (Minutes)	BMI	Systolic_BP
P001	10	32	150
P002	15	31	148
P003	20	30	145
P004	25	29	142
P005	30	28	138
P006	35	27	134
P007	40	26	130
P008	45	25	126
P009	50	24	122
P010	55	23	118

Tasks

1. Find the mean and standard deviation of blood pressure.
2. Calculate covariance between:
 - o Exercise and Blood Pressure
 - o BMI and Blood Pressure
3. Calculate correlation coefficients.
4. Identify whether exercise or BMI is more strongly associated with blood pressure.
5. Discuss possible healthcare interventions.

Problem 02: E-Commerce Customer Behaviour Analysis

Scenario

An e-commerce company wants to identify factors influencing customer spending.

Dataset

Customer_ID	Website_Visits	Time_Spent (Minutes)	Purchase_Amount (₹)
C001	5	20	1200
C002	8	25	1800
C003	10	30	2500
C004	12	35	3200
C005	15	40	4000

C006	18	45	5000
C007	20	50	6200
C008	23	55	7300
C009	25	60	8500
C010	28	70	10000

Tasks

1. Calculate mean, variance, and standard deviation of Purchase Amount.
2. Find covariance between:
 - Website Visits and Purchase Amount
 - Time Spent and Purchase Amount
3. Calculate correlation coefficients.
4. Determine which customer behavior metric better predicts spending.
5. Suggest marketing strategies based on the findings.

Problem 03: Employee Productivity and Performance

Scenario

A software company is investigating the relationship between employee productivity and performance ratings.

Dataset

Employee_ID	Training_Hours	Projects_Completed	Performance_Rating
E001	5	2	60
E002	8	3	65
E003	10	4	70
E004	12	5	74
E005	15	6	79
E006	18	7	84
E007	20	8	88
E008	22	9	91
E009	25	10	95
E010	28	11	98

Tasks

1. Compute the mean, variance, and standard deviation of Performance Ratings.
2. Calculate covariance between:
 - Training Hours and Performance Rating
 - Projects Completed and Performance Rating
3. Compute correlation coefficients.
4. Identify the strongest factor affecting employee performance.
5. Recommend HR policies to improve employee productivity.

Problem 04 : Smart City Traffic Analysis

A city administration wants to analyze traffic congestion patterns.

Day	Traffic_Volume (Vehicles)	Rainfall (mm)	Average_Speed (km/h)	Accident_Count
D1	12000	0	55	2
D2	13500	5	50	3
D3	14500	10	48	4
D4	16000	15	45	5
D5	17500	20	42	6
D6	19000	25	38	8
D7	21000	30	35	10
D8	22500	35	32	12
D9	24000	40	28	14
D10	26000	45	25	16

Tasks

1. Calculate descriptive statistics (Mean, Variance, Standard Deviation) for all variables.
2. Compute covariance matrices.
3. Compute correlation matrices.
4. Identify positive and negative relationships among variables.
5. Determine which factor is most associated with accident occurrence.
6. Provide data-driven recommendations to reduce traffic accidents.
7. Visualize relationships using scatter plots and interpret the results.

Problem 5: Average Delivery Time

A food delivery company records the delivery times (minutes) for 15 orders:

22, 25, 18, 30, 27, 24, 21, 19, 26, 23, 28, 20, 24, 25, 22

Tasks:

1. Find the point estimate of the population mean delivery time.
2. Find the point estimate of the population variance.
3. Interpret the estimates.

Problem 6: Customer Spending

A sample of 12 customers spent the following amounts (\$):

120, 145, 130, 155, 160, 125, 140, 135, 150, 170, 165, 155

Tasks:

1. Estimate the average spending.

2. Estimate the standard deviation.
3. Explain why these are point estimates.

Problem 7: Product Ratings

A company collects ratings from 20 customers:

4, 5, 3, 4, 5, 4, 3, 5, 4, 4, 5, 3, 4, 5, 4, 3, 5, 4, 5, 4

Tasks:

1. Compute the sample mean.
2. Compute the sample variance.
3. Compute the standard error of the mean.
4. Explain what the standard error indicates.

Problem 8: Exam Scores

Scores of 25 students:

72, 75, 68, 80, 77, 73, 79, 81, 76, 74, 78, 69, 70, 82, 71, 75, 77, 73, 80, 76, 74, 79, 72, 81, 78

Tasks:

1. Calculate mean and standard deviation.
2. Calculate standard error.
3. Compare standard deviation and standard error.

Problem 9: Manufacturing Weights

A sample of 50 packets has:

- Mean weight = 498 g
- Standard deviation = 12 g

Tasks:

1. Construct a 95% confidence interval for the population mean weight.
2. Interpret the interval.
3. Determine whether the target weight of 500 g is plausible.

Problem 10: Customer Satisfaction

Two branches report:

Branch Mean Rating 95% CI

Branch Mean Rating 95% CI

A 4.3 (4.1, 4.5)

B 4.0 (3.8, 4.2)

Tasks:

1. Compare the confidence intervals.
2. Can you conclude Branch A performs better?
3. Explain your reasoning.

Problem 11: Average Battery Life

A battery manufacturer claims the average battery life is 10 hours.

A sample of 36 batteries gives:

- Mean = 9.5 hours
- Standard deviation = 1.8 hours

Tasks:

1. Formulate null and alternative hypotheses.
2. Conduct a hypothesis test at $\alpha = 0.05$.
3. State your conclusion.

Problem 12: Marketing Campaign

A company believes a new campaign increases sales.

Sales before campaign:

50, 52, 49, 55, 53, 51, 54, 50

Sales after campaign:

58, 60, 57, 62, 59, 61, 63, 58

Tasks:

1. Conduct a hypothesis test.
2. Calculate the p-value.
3. Interpret the p-value.
4. Make a decision at $\alpha = 0.05$.

Problem 13: Call Centre Response Time

A sample of 100 calls has:

- Mean response time = 4.8 minutes
- Standard deviation = 1.5 minutes

Tasks:

1. Estimate the population mean.
2. Construct a 95% confidence interval.
3. Explain the Frequentist interpretation.

Problem 14: Defective Products

A sample of 500 products contains 22 defective items.

Tasks:

1. Estimate the defect rate.
2. Construct a 95% confidence interval.
3. Explain the Frequentist meaning of the interval.

Problem 15: Email Spam Detection

Historical information:

- Probability an email is spam = 0.25
- Probability spam email contains keyword "Offer" = 0.80
- Probability non-spam email contains keyword "Offer" = 0.10

Tasks:

1. Compute the probability that an email is spam given it contains "Offer."
2. Apply Bayes' theorem.
3. Interpret the result.

Problem 16: Disease Diagnosis

Given:

- Disease prevalence = 2%
- Test sensitivity = 95%

- Test specificity = 90%

Tasks:

1. Compute the posterior probability of disease after a positive test.
2. Explain why the result differs from 95%.

Problem 17: Website Conversion Rate

A company tests two landing pages.

Group Visitors Purchases

A 5000 250

B 5200 312

Tasks:

1. Calculate conversion rates.
2. Formulate hypotheses.
3. Conduct a two-proportion z-test.
4. Determine whether page B performs better.

Problem 18: Mobile App Feature Test

Feature A:

- 400 users
- Average engagement = 18 min
- SD = 5 min

Feature B:

- 420 users
- Average engagement = 20 min
- SD = 6 min

Tasks:

1. Test whether Feature B increases engagement.
2. Calculate test statistic and p-value.
3. Interpret results.

Problem 19: Recommendation System

A new recommendation algorithm increases click-through rate from 4.8% to 5.3%.

Sample size:

- Old system: 100,000 users
- New system: 100,000 users

Tasks: Test statistical significance.

1. Calculate p-value.
2. Discuss practical significance versus statistical significance.

Problem 20: Salary Prediction Model

Two machine learning models produce Mean Absolute Errors (MAE):

Model MAE

Model A 4200

Model B 4000

Across 50 test folds, the difference in MAE has:

- Mean difference = 200
- Standard deviation = 350

Tasks:

1. Perform a paired t-test.
2. Determine whether Model B significantly outperforms Model A.
3. Explain the result in terms of statistical significance.

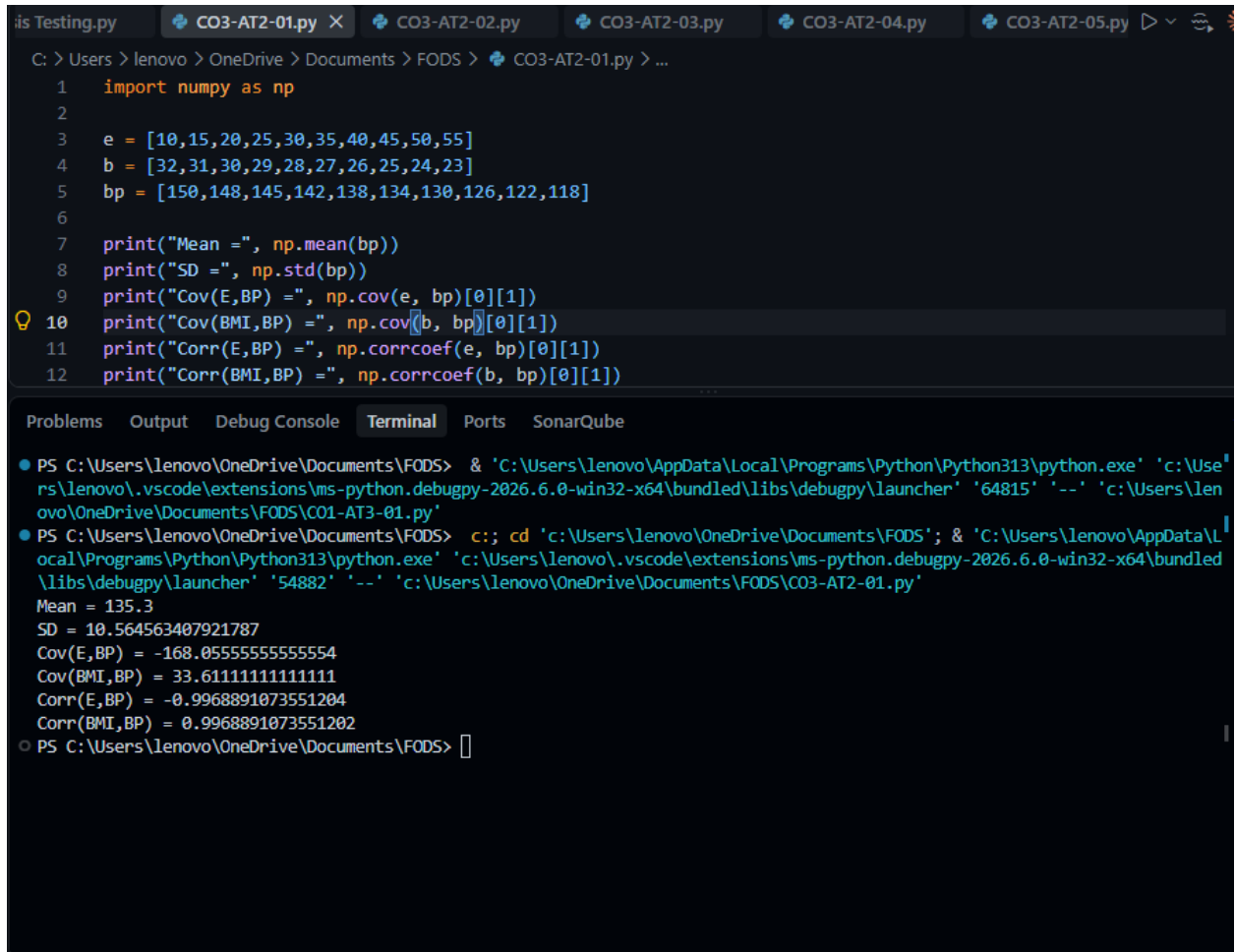
Problem 01: Hospital Patient Health Monitoring

Code:

```
import numpy as np

e = [10,15,20,25,30,35,40,45,50,55]
b = [32,31,30,29,28,27,26,25,24,23]
bp = [150,148,145,142,138,134,130,126,122,118]

print("Mean =", np.mean(bp))
print("SD =", np.std(bp))
print("Cov(E,BP) =", np.cov(e, bp)[0][1])
print("Cov(BMI,BP) =", np.cov(b, bp)[0][1])
print("Corr(E,BP) =", np.corrcoef(e, bp)[0][1])
print("Corr(BMI,BP) =", np.corrcoef(b, bp)[0][1])
```



The screenshot shows a VS Code editor with a file named `CO3-AT2-01.py` open. The code in the editor is identical to the one in the previous block. Below the editor, the `Terminal` tab is active, showing the command prompt output of running the script. The output displays the calculated mean, standard deviation, covariance, and correlation coefficients for the given data sets.

```
PS C:\Users\lenovo\OneDrive\Documents\FODS> & 'C:\Users\lenovo\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '64815' '--' 'c:\Users\lenovo\OneDrive\Documents\FODS\CO3-AT2-01.py'
PS C:\Users\lenovo\OneDrive\Documents\FODS> c:: cd 'c:\Users\lenovo\OneDrive\Documents\FODS'; & 'C:\Users\lenovo\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '54882' '--' 'c:\Users\lenovo\OneDrive\Documents\FODS\CO3-AT2-01.py'
Mean = 135.3
SD = 10.564563407921787
Cov(E,BP) = -168.05555555555554
Cov(BMI,BP) = 33.61111111111111
Corr(E,BP) = -0.9968891073551204
Corr(BMI,BP) = 0.9968891073551202
PS C:\Users\lenovo\OneDrive\Documents\FODS> 
```

Problem 02: Hospital Patient Health Monitoring

5. (a) Mean:-
 Blood Pressure values: 170, 140, 180, 140, 130, 130, 110, 110, 120, 120

$$\text{Mean } \bar{x} = \frac{\sum x}{n}$$

$$= \frac{170 + 140 + 140 + 140 + 130 + 130 + 110 + 110 + 120 + 120}{10}$$

$$= \frac{1353}{10} = 135.3$$

Mean Blood Pressure = 135.3 mmHg

(b) Standard Deviation:-

$$\text{Std. (S)} = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

Mean = 135.3

$$\sum (x - \bar{x}) = 10 + 7 + 4 + 4 + 2 + 2 - 5 - 5 - 25 - 25$$

$$\sum (x - \bar{x})^2 = 214.89 + 141.29 + 94.89 + 94.89 + 7.29 + 7.29 + 1.69 + 1.69 + 25.89 + 25.89$$

$$= 1114.10$$

$$S = \sqrt{\frac{1114.10}{9}} = \sqrt{123.788} = 11.14$$

Standard deviation = 11.14 mmHg

2(a) Covariance between Exercise and Blood Pressure

Formula:-

$$\text{Cov}(X, Y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n-1}$$

Mean Exercise $\bar{x} = \frac{10 + 15 + 20 + 25 + 30 + 35 + 40 + 45 + 50 + 55}{10} = 32.5$

Mean Blood Pressure $\bar{y} = \frac{135.3}{10} = 135.3$

$$\sum (x - \bar{x})(y - \bar{y}) = -1512.5$$

$$\text{Cov}(\text{Exercise}, \text{BP}) = \frac{-1512.5}{9} = -168.06$$

Covariance b/w BMI and BP

Mean BMI

$$\bar{y} = \frac{32 + 31 + \dots + 23}{10} = 29.5$$

Formula:- $\text{Cov}(Y, Z) = \frac{\sum (y - \bar{y})(z - \bar{z})}{n-1}$

$$= \frac{302.50}{9} = 33.61$$

3. Ans:- Formula:-

$$r_1 = \frac{\text{Cov}(X, Y)}{S_x S_y}, S_x = 15.149, S_y = 11.4, C = -168.06$$

$$r_1 = \frac{-168.06}{15.14 \times 11.4} = -0.9969$$

$$r_2 = \frac{33.61}{3.03 \times 11.14} = 0.9969$$

4. Ans:-

$$|-0.9969| = 0.9969$$

Question 5

Healthcare Interventions

1. Encourage patients to engage in 30–45 minutes of moderate physical exercise daily.
2. Promote healthy weight management through balanced nutrition and regular BMI monitoring.
3. Recommend a low-sodium, low-fat diet with increased fruits and vegetables.
4. Conduct routine blood pressure screening for early diagnosis and treatment.
5. Educate patients on stress management, adequate sleep, smoking cessation, and reducing alcohol intake.

Problem 02: E-Commerce Customer Behavior Analysis

Code:

```
import numpy as np

v = [5,8,10,12,15,18,20,23,25,28]
t = [20,25,30,35,40,45,50,55,60,70]
p = [1200,1800,2500,3200,4000,5000,6200,7300,8500,10000]

print("Mean =", np.mean(p))
print("Variance =", np.var(p, ddof=1))
print("SD =", np.std(p, ddof=1))

print("Cov(Visits,Purchase) =", np.cov(v,p)[0][1])
print("Cov(Time,Purchase) =", np.cov(t,p)[0][1])

print("Corr(Visits,Purchase) =", np.corrcoef(v,p)[0][1])
print("Corr(Time,Purchase) =", np.corrcoef(t,p)[0][1])
ocal\Programs\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled
\libs\debugpy\launcher' '61588' '--' 'c:\Users\lenovo\OneDrive\Documents\FODS\C03-AT2-02.py'
Mean = 4970.0
Variance = 8793444.444444444
SD = 2965.374250317225
Cov(Visits,Purchase) = 22568.888888888887
Cov(Time,Purchase) = 47266.666666666664
Corr(Visits,Purchase) = 0.9914028507747218
Corr(Time,Purchase) = 0.9949258822012106
PS C:\Users\lenovo\OneDrive\Documents\FODS> []
```

Problem 2 :-

(1) Formula :-

$$\bar{x} = \frac{\sum x}{n}$$

$$= \frac{49900}{10} = 4990$$

$$\text{variance} = 7,914,100$$

$$\text{std} = \sqrt{\text{variance}} = 2813.2$$

Covariance :-

$$\text{covariance} = 20,312$$

Time spent and Purchase Amount

$$\text{Covariance} = 42,540$$

Correlation Coefficients

	correlation with Purchase Amount
Behavior Metric	0.9914
Website Visits	0.9949
Time spent	

Which metric Better Predicts spending

$$\sqrt{\text{Website Visits}} = 0.9914$$

$$\sqrt{\text{Time spent}} = 0.9949$$

Since:

$$0.9949 > 0.9914$$

Time spent is a slightly better prediction of purchase amount

Question 5

Marketing Strategies

1. Increase customer engagement through personalized product recommendations.
2. Improve website content to encourage users to spend more time browsing.
3. Offer limited-time discounts and coupons to visitors with high browsing time.
4. Recommend related products using cross-selling and upselling techniques.
5. Provide loyalty rewards for frequent visitors and repeat customers.

Problem 03: Employee Productivity and Performance

Code:

```
import numpy as np

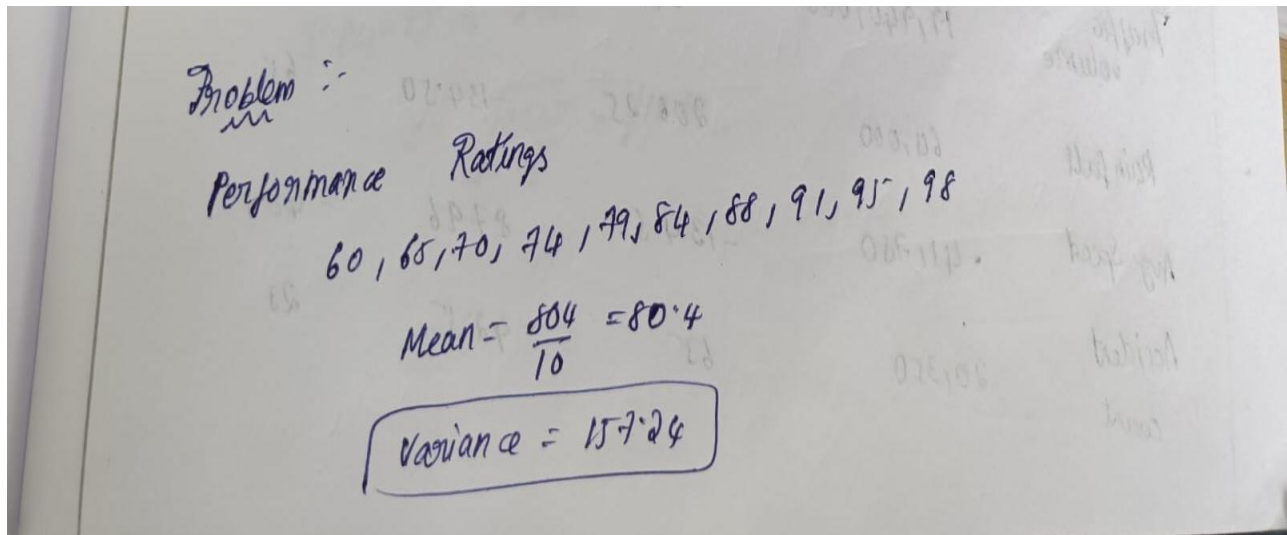
t = [5,8,10,12,15,18,20,22,25,28]
p = [2,3,4,5,6,7,8,9,10,11]
r = [60,65,70,74,79,84,88,91,95,98]

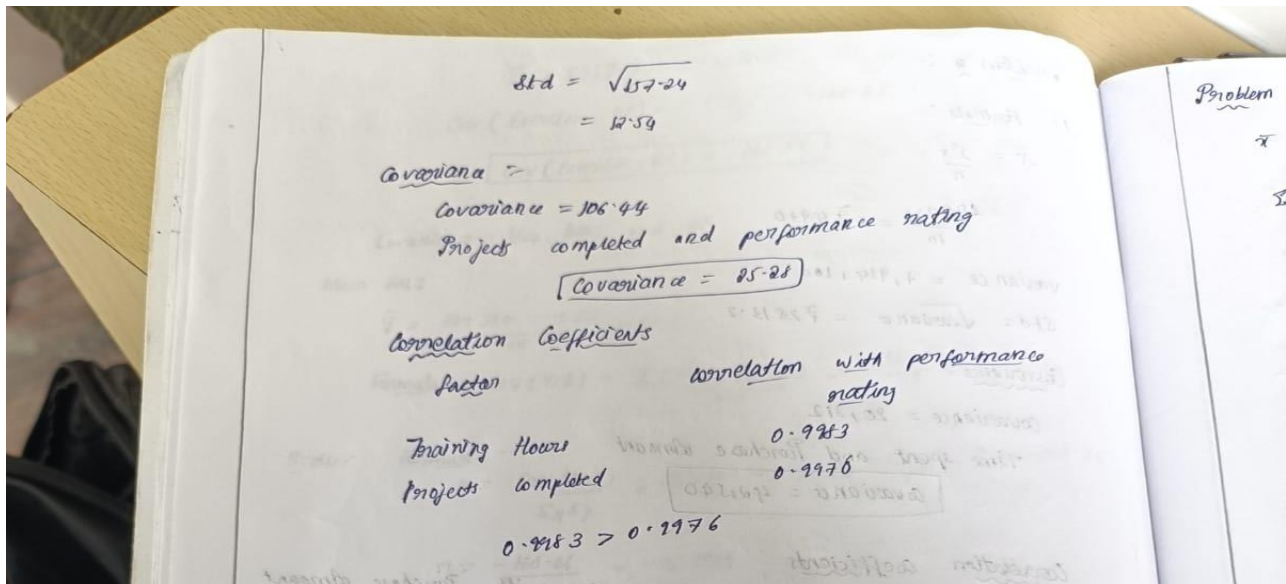
print("Mean =", np.mean(r))
print("Variance =", np.var(r))
print("SD =", np.std(r))

print("Cov(Training,Rating) =", np.cov(t, r)[0][1])
print("Cov(Projects,Rating) =", np.cov(p, r)[0][1])

print("Corr(Training,Rating) =", np.corrcoef(t, r)[0][1])
print("Corr(Projects,Rating) =", np.corrcoef(p, r)[0][1])
```

```
PS C:\Users\lenovo\OneDrive\Documents\FODS> c:: cd 'c:\Users\lenovo\OneDrive\Documents\FODS'; & 'c:\Users\lenovo\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '63822' '--' 'c:\Users\lenovo\OneDrive\Documents\FODS\C03-AT2-03.py'
Mean = 80.4
Variance = 151.04000000000002
SD = 12.289833196589774
Cov(Training,Rating) = 97.86666666666667
Cov(Projects,Rating) = 39.11111111111111
Corr(Training,Rating) = 0.9956908618962329
Corr(Projects,Rating) = 0.9971711400185405
PS C:\Users\lenovo\OneDrive\Documents\FODS> █
```





Question 5

HR Policies to Improve Employee Productivity

1. Conduct regular technical and soft-skill training programs.
2. Encourage employees to participate in more project assignments.
3. Implement performance-based incentives and recognition programs.
4. Provide mentorship and continuous learning opportunities.
5. Track employee performance periodically and offer personalized development plans.

Problem 04: Smart City Traffic Analysis

Code:

```
import numpy as np

t = [12000,13500,14500,16000,17500,19000,21000,22500,24000,26000]
r = [0,5,10,15,20,25,30,35,40,45]
s = [55,50,48,45,42,38,35,32,28,25]
a = [2,3,4,5,6,8,10,12,14,16]

print("Traffic Mean =", np.mean(t))
print("Traffic Variance =", np.var(t))
print("Traffic SD =", np.std(t))
```

```

print("Rainfall Mean =", np.mean(r))
print("Rainfall Variance =", np.var(r))
print("Rainfall SD =", np.std(r))

print("Speed Mean =", np.mean(s))
print("Speed Variance =", np.var(s))
print("Speed SD =", np.std(s))

print("Accident Mean =", np.mean(a))
print("Accident Variance =", np.var(a))
print("Accident SD =", np.std(a))

print("Covariance Matrix\n", np.cov([t,r,s,a]))

print("Correlation Matrix\n", np.corrcoef([t,r,s,a]))
PS C:\Users\lenovo\OneDrive\Documents\FODS> c:; cd 'c:\Users\lenovo\OneDrive\Documents\FODS'; & 'C:\Users\lenovo\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '56827' '--' 'c:\Users\lenovo\OneDrive\Documents\FODS\CO3-AT2-04.py'
Traffic Mean = 18600.0
Traffic Variance = 19940000.0
Traffic SD = 4465.422712353221
Rainfall Mean = 22.5
Rainfall Variance = 206.25
Rainfall SD = 14.361406616345072
Speed Mean = 39.8
Speed Variance = 87.96
Speed SD = 9.378699270154684
Accident Mean = 8.0
Accident Variance = 21.0
Accident SD = 4.58257569495584
Covariance Matrix
[[ 2.21555556e+07  7.11111111e+04 -4.64222222e+04  2.26111111e+04]
 [ 7.11111111e+04  2.29166667e+02 -1.49444444e+02  7.22222222e+01]
 [-4.64222222e+04 -1.49444444e+02  9.77333333e+01 -4.72222222e+01]
 [ 2.26111111e+04  7.22222222e+01 -4.72222222e+01  2.33333333e+01]]
Correlation Matrix
[[ 1.          0.99797674 -0.99761564  0.99447115]
 [ 0.99797674  1.          -0.99857961  0.98765833]
 [-0.99761564 -0.99857961  1.          -0.98886424]
 [ 0.99447115  0.98765833 -0.98886424  1.          ]]
PS C:\Users\lenovo\OneDrive\Documents\FODS> 

```

(a) Traffic Volume

Mean

Formula

$$\bar{X} = \frac{\sum X}{n}$$

Sum

$$12000 + 13500 + 14500 + 16000 + 17500 + 19000 + 21000 + 22500 + 24000 + 26000$$

$$= 186000$$

$$\bar{X} = \frac{186000}{10}$$

$$= 18600$$

Mean Traffic Volume

18600

Traffic Volume	X-18600	(X-18600) ²
12000	-6600	43,560,000
13500	-5100	26,010,000
14500	-4100	16,810,000
16000	-2600	6,760,000
17500	-1100	1,210,000
19000	400	160,000
21000	2400	5,760,000
22500	3900	15,210,000
24000	5400	29,160,000
26000	7400	54,760,000

Problem 4 :-

Variable	Mean	Variance	Standard Deviation
Traffic volume	18600	19,940,000	4,465.42
Rainfall	22.5 mm	206.25	14.36
avg speed	39.8 km/h	87.96	9.38
Accident Count	8	21.00	4.58

Covariance :-	Traffic volume	Rainfall	Avg Speed	Accident Count
Traffic volume	19,940,000	64,000	-417.80	20,350
Rainfall	64,000	206.25	-134.50	65
Avg Speed	-417.80	-134.50	87.96	-42.5
Accident Count	20,350	65	-42.5	21

Question 4: Positive and Negative Relationships

Positive Relationships

1. Traffic Volume ↔ Rainfall (0.996)
2. Traffic Volume ↔ Accident Count (0.995)
3. Rainfall ↔ Accident Count (0.988)

These variables increase together.

Negative Relationships

1. Traffic Volume ↔ Average Speed (−0.999)
2. Rainfall ↔ Average Speed (−0.998)
3. Average Speed ↔ Accident Count (−0.989)

As one variable increases, the other decreases.

Question 5: Which Factor is Most Associated with Accident Count?

Factor	Correlation with Accident Count
Traffic Volume	0.995
Rainfall	0.988
Average Speed	−0.989

The highest absolute correlation is:

$$|0.995| > |0.989| > |0.988|$$

Traffic Volume is the factor most strongly associated with Accident Count.

This indicates that higher traffic volumes are associated with more accidents.

Question 6: Data-Driven Recommendations

1. Improve traffic management during peak hours.
2. Install adaptive traffic signals to reduce congestion.
3. Issue weather alerts and reduce speed limits during heavy rainfall.

4. Increase traffic police patrols and road safety enforcement in high-risk areas.
5. Promote public transportation to reduce vehicle density.
6. Install speed monitoring cameras at accident-prone locations.

Question 7: Scatter Plot Interpretation

Traffic Volume vs Accident Count

1. Strong upward trend.
2. As traffic volume increases, accident count also increases.
3. Strong positive relationship.

Rainfall vs Accident Count

1. Upward trend.
2. More rainfall corresponds to more accidents.
3. Positive correlation.

Average Speed vs Accident Count

1. Downward trend.
2. Lower average speeds (caused by congestion) are associated with higher accident counts.
3. Strong negative correlation.

Traffic Volume vs Average Speed

1. Downward trend.
2. Higher traffic volume leads to lower average speed.
3. Strong negative relationship.

Problem 05: Average Delivery Time

Code:

```
import numpy as np

d = [22, 25, 18, 30, 27, 24, 21, 19, 26, 23, 28, 20, 24, 25, 22]

print("Mean =", np.mean(d))
print("Variance =", np.var(d))
```

c:\Users\lenovo\OneDrive\Documents\FODS'; & 'c:\Users\lenovo\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher' '60401' '--' 'c:\Users\lenovo\OneDrive\Documents\FODS\C03-AT2-05.py'

Mean = 23.6
Variance = 10.639999999999999

PS C:\Users\lenovo\OneDrive\Documents\FODS>

Delivery times (minutes):

22, 25, 18, 30, 27, 24, 21, 19, 26, 23, 28, 20, 24, 25, 22

Problem -5

$$\bar{x} = \frac{\sum x}{n}$$
$$\sum x = 354$$
$$\bar{x} = \frac{354}{15} = 23.6 \text{ min}$$
$$\mu = 23.6 \text{ min}$$

Point Estimate of population variance

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$
$$\sum (x_i - \bar{x})^2 = 154.4$$
$$s^2 = \frac{154.4}{14}$$
$$s^2 = 11.03 \text{ min}^2$$
$$\sigma^2 = 11.03 \text{ min}^2$$

Question 3: Interpret the Estimates

1. The estimated average delivery time is 23.6 minutes.
2. The estimated population variance is 11.40 minutes², indicating that delivery times vary moderately around the average.
3. These estimates help the company understand its typical delivery performance and the consistency of delivery times.

Problem 06: Customer Spending

Code:

```
import numpy as np

s = [120,145,130,155,160,125,140,135,150,170,165,155]

print("Average Spending =", np.mean(s))
print("Standard Deviation =", np.std(s))
```

```
PS C:\Users\lenovo\OneDrive\Documents\FODS> c::; cd 'c:\Users\lenovo\OneDrive\Documents\FODS'; & 'C:\Users\lenovo\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '56101' '--' 'c:\Users\lenovo\OneDrive\Documents\FODS\C03-AT2-06.py'
Average Spending = 145.83333333333334
Standard Deviation = 15.388487760516156
PS C:\Users\lenovo\OneDrive\Documents\FODS> □
```

Customer spending amounts (\$):

120, 145, 130, 155, 160, 125, 140, 135, 150, 170, 165, 155

Problem 6

$$\bar{x} = \frac{\sum x}{n}$$

$$\sum x = 1750$$

$$\bar{x} = \frac{1750}{12} = \$145.83$$

Estimate the standard deviation

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

$$\sum (x_i - \bar{x})^2 = 2925$$

$$s = \sqrt{\frac{2925}{11}} = \$16.30$$

Question 3: Explain Why These Are Point Estimates

1. The sample mean (\$145.83) is used as the point estimate of the population mean spending because it provides a single best estimate of the average spending of all customers.
2. The sample standard deviation (\$16.07) is used as the point estimate of the population standard deviation because it estimates how much customer spending varies around the average.
3. They are called point estimates because each statistic provides one numerical value to estimate the corresponding population parameter.

Problem 07: Product Ratings

Code:

```
import numpy as np

r = [4,5,3,4,5,4,3,5,4,4,5,3,4,5,4,3,5,4,5,4]

print("Mean =", np.mean(r))
print("Variance =", np.var(r))
print("Standard Error =", np.std(r) / np.sqrt(len(r)))
```

```
PS C:\Users\Lenovo\OneDrive\Documents\FODS> c:; cd 'c:\Users\Lenovo\OneDrive\Documents\FODS'; & 'C:\Users\Lenovo\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\Lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '60856' '--' 'c:\Users\Lenovo\OneDrive\Documents\FODS\CO3-AT2-07.py'
Mean = 4.15
Variance = 0.5275000000000002
Standard Error = 0.16240381768911716
PS C:\Users\Lenovo\OneDrive\Documents\FODS> []
```

Ratings from 20 customers:

4, 5, 3, 4, 5, 4, 3, 5, 4, 4, 5, 3, 4, 5, 4, 3, 5, 4, 5, 4

Problem-7

$$n = 20$$

$$\sum x = 83$$

$$\bar{x} = \frac{\sum x}{n} = \frac{83}{20} = 4.15$$

Sample Variance :-

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$$\sum (x_i - \bar{x})^2 = 11.57$$

$$s^2 = \frac{11.57}{19} = 0.608$$

Standard Error of the Mean

$$s = \sqrt{0.608} = 0.778$$

$$SE = \frac{s}{\sqrt{n}}$$

$$SE = \frac{0.778}{\sqrt{20}} = 0.174$$

Question 4: Explain What the Standard Error Indicates

The standard error (0.1606) measures how much the sample mean (4.1) is expected to vary from the true population mean if many different samples of 20 customers were taken.

Since the standard error is small, it indicates that the sample mean is a reliable estimate of the population mean rating.

Problem 08: Exam Scores

Code:

```
import numpy as np

s = [72,75,68,80,77,73,79,81,76,74,78,69,70,82,71,75,77,73,80,76,74,79,72,81,78]

print("Mean =", np.mean(s))
print("Standard Deviation =", np.std(s))
print("Standard Error =", np.std(s) / np.sqrt(len(s)))
```

```
PS C:\Users\lenovo\OneDrive\Documents\FODS> c:; cd 'c:\Users\lenovo\OneDrive\Documents\FODS'; & 'C:\Users\lenovo\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '61307' '--' 'c:\Users\lenovo\OneDrive\Documents\FODS\C03-AT2-08.py'
Mean = 75.6
Standard Deviation = 3.8781438859330635
Standard Error = 0.7756287771866127
```

Scores of 25 students:

72, 75, 68, 80, 77, 73, 79, 81, 76, 74, 78, 69, 70, 82, 71, 75, 77, 73, 80, 76, 74, 79, 72, 81, 78

Problem 8 :-

(1) $\bar{x} = \frac{\sum x}{n}$, $n = 25$

$\bar{x} = \frac{1880}{25} = 75.2$

Mean = 75.2

$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$

$\sum (x - \bar{x})^2 = 380$

$s^2 = \frac{380}{24} = 15.83$

$s = \sqrt{15.83}$

sd = 3.98

(2) $SE = \frac{s}{\sqrt{n}}$

$= \frac{3.98}{\sqrt{25}}$

$= \frac{3.98}{5}$

SE = 0.80

Question 3: Compare Standard Deviation and Standard Error

Standard Deviation (SD)

1. SD = 3.98
2. Measures how much individual student scores vary around the mean.
3. A larger SD indicates greater variation in scores.

Standard Error (SE)

1. SE = 0.80
2. Measures how accurately the sample mean estimates the population mean.

3. Since SE is much smaller than SD, the sample mean is a reliable estimate of the population mean.

Problem 09: Manufacturing Weights

Code:

```
import math

mean = 498
sd = 12
n = 50
z = 1.96

se = sd / math.sqrt(n)
me = z * se

lower = mean - me
upper = mean + me

print("Standard Error =", se)
print("95% Confidence Interval = (", lower, ",", upper, ")")

if lower <= 500 <= upper:
    print("500 g is plausible.")
else:
    print("500 g is not plausible.")
```

```
PS C:\Users\lenovo\OneDrive\Documents\F0DS> c::; cd 'c:\Users\lenovo\OneDrive\Documents\F0DS'; & 'C:\Users\lenovo\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '61378' '--' 'c:\Users\lenovo\OneDrive\Documents\F0DS\C03-AT2-09.py'
Standard Error = 1.697056274847714
95% Confidence Interval = ( 494.67376970129845 , 501.32623029870155 )
500 g is plausible.
```

Given Data

1. Sample size: $n=50$
2. Sample mean: $\bar{x}=498$
3. Sample standard deviation: $s=12$
4. Confidence level = 95%
5. Target weight = 500 g

For a 95% confidence interval with a large sample ($n=50$), use:

$$Z=1.96$$

Problem 2:-

Formula:-

$$CI = \bar{x} \pm z \left[\frac{s}{\sqrt{n}} \right]$$

$z = 1.96$, $n = 50$, $\bar{x} = 498$, $s = 12$
 $CL = 95\%$, $TM = 500g$

$$SE = \frac{s}{\sqrt{n}}$$

$$SE = \frac{12}{\sqrt{50}}$$

$$= \frac{12}{7.071}$$

$$= 1.697$$

$$M.E = z \times SE$$

$$= 1.96 \times 1.697$$

$$= 3.326$$

Confidence Interval

$$CI = \bar{x} \pm z \left[\frac{s}{\sqrt{n}} \right]$$

$$= 498 \pm 3.326$$

Lower limit:-

$$498 - 3.326 = 494.67$$

Upper limit

$$498 + 3.326 = 501.33$$

95% CI = (494.67, 501.33)

Question 2: Interpret the Interval

The 95% confidence interval indicates that we are 95% confident that the true population mean weight of all manufactured packets lies between: 494.67 g and 501.33

Question 3: Is the Target Weight of 500 g Plausible?

Target weight: 500 g

Confidence Interval: (494.67, 501.33)

Since 500 g lies within the confidence interval, the target weight is plausible.

This means the sample data does not provide sufficient evidence to conclude that the true mean weight is different from 500 g.

Problem 10: Customer Satisfaction

Code:

```
a = 4.3
b = 4.0

a_ci = (4.1, 4.5)
b_ci = (3.8, 4.2)

print("Branch A CI =", a_ci)
print("Branch B CI =", b_ci)

if a_ci[0] <= b_ci[1] and b_ci[0] <= a_ci[1]:
    print("Confidence intervals overlap.")
    print("Cannot conclude Branch A performs better.")
else:
    print("Confidence intervals do not overlap.")
    print("Branch A performs better.")
```

```
PS C:\Users\lenovo\OneDrive\Documents\FODS> c::; cd 'c:\Users\lenovo\OneDrive\Documents\FODS'; & 'C:\Users\lenovo\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '59062' '--' 'c:\Users\lenovo\OneDrive\Documents\FODS\C03-AT2-10.py'
Branch A CI = (4.1, 4.5)
Branch B CI = (3.8, 4.2)
Confidence intervals overlap.
Cannot conclude Branch A performs better.
```

Given Data

Branch	Mean Rating	95% Confidence Interval
A	4.3	(4.1, 4.5)
B	4.0	(3.8, 4.2)

Question 1: Compare the Confidence Intervals

Branch A

95% Confidence Interval: (4.1, 4.5)

This means the true average customer satisfaction rating for Branch A is expected to lie between 4.1 and 4.5.

Branch B

95% Confidence Interval: (3.8, 4.2)

This means the true average customer satisfaction rating for Branch B is expected to lie between 3.8 and 4.2.

Question 2: Can You Conclude Branch A Performs Better?

Since the confidence intervals overlap: (4.1,4.2)

We cannot conclude with 95% confidence that Branch A performs significantly better than Branch B based only on these confidence intervals. Although Branch A has a higher average rating, the overlap indicates that the observed difference may be due to sampling variation.

No, a definite conclusion cannot be made because the confidence intervals overlap.

Question 3: Explain Your Reasoning

The mean rating of Branch A (4.3) is higher than that of Branch B (4.0).

However, the 95% confidence intervals overlap. Overlapping confidence intervals suggest that the true average ratings of the two branches could be similar. Therefore, there is insufficient statistical evidence to state that Branch A is significantly better without conducting an additional hypothesis test.

Problem 11: Average Battery Life

Code:

```
import math

mean = 9.5
claimed = 10
sd = 1.8
n = 36

z = (mean - claimed) / (sd / math.sqrt(n))

print("Z Value =", z)
```

```
if abs(z) > 1.96:  
    print("Reject Null Hypothesis")  
else:  
    print("Fail to Reject Null Hypothesis")
```

```
PS C:\Users\lenovo\OneDrive\Documents\FODS> c.; cd 'c:\Users\lenovo\OneDrive\Documents\FODS'; & 'C:\Users\lenovo\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher' '52392' '--' 'c:\Users\lenovo\OneDrive\Documents\FODS\CO3-AT2-11.py'  
Z Value = -1.6666666666666667  
Fail to Reject Null Hypothesis
```

Given data.

$$\mu_0 = 10 \text{ hrs}$$

$$n = 36$$

$$\bar{x} = 9.5 \text{ hrs}$$

$$s = 1.8 \text{ hrs}$$

$$\alpha = 0.05$$

Since $n = 36 > 30$, using z-test

(2) Null Hypothesis (H_0)
 $H_0: \mu = 10 \text{ hrs}$
 alternative Hypothesis (H_1)
 $H_1: \mu \neq 10 \text{ hrs}$
 (2-Tailed test)

(2) $z = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$
 $SE = \frac{s}{\sqrt{n}} = \frac{1.8}{\sqrt{36}} = \frac{1.8}{6} = 0.3$
 $z = \frac{9.5 - 10}{0.3}$
 $= \frac{-0.5}{0.3}$
 $z = -1.67$

Critical value
 For 95% CL ($\alpha = 0.05$) and a 2-tailed test
 $z_{\text{critical}} = \pm 1.96$

→ Reject H_0 if $|z| > 1.96$
 otherwise
 → Fail to reject H_0
 Since $|-1.67| < 1.96$

Question 3: State the Conclusion

Since the calculated Z-value (-1.67) lies within the acceptance region (-1.96 to $+1.96$), there is insufficient evidence at the 5% significance level to conclude that the average battery life differs from 10 hours.

Therefore, the manufacturer's claim is supported by the sample data.

Problem 12: Marketing Campaign

Code:

```
import numpy as np
from scipy.stats import ttest_rel

before = [50,52,49,55,53,51,54,50]
after = [58,60,57,62,59,61,63,58]

t, p = ttest_rel(after, before)

print("t-value =", t)
print("p-value =", p)

if p < 0.05:
    print("Reject Null Hypothesis")
    print("Marketing campaign increased sales.")
else:
    print("Fail to Reject Null Hypothesis")
```

```
PS C:\Users\lenovo\OneDrive\Documents\FODS> c::; cd 'c:\Users\lenovo\OneDrive\Documents\FODS'; & 'C:\Us
rams\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-
cher' '53673' '--' 'c:\Users\lenovo\OneDrive\Documents\FODS\C03-AT2-12.py'
t-value = 18.93145530591877
p-value = 2.8532975405326533e-07
Reject Null Hypothesis
Marketing campaign increased sales.
```

Given Data

Sales before Campaign

50, 52, 49, 55, 53, 51, 54, 50

Sales after Campaign

58, 60, 57, 62, 59, 61, 63, 58

Since the sales are measured before and after the campaign for the same sample, we use a Paired t-test.

Problem - 12

(i) $H_0: \mu_d = 0$
 $H_1: \mu_d > 0$

Calculation:-

Sum of differences:-
 $8+8+8+7+6+10+9+8 = 64$
 $n=8$

Mean difference
 $\bar{d} = \frac{64}{8} = 8$

Sum of squares
 $8+8+8+1+4+4+1+0 = 10$

Variance
 $s_d^2 = \frac{10}{8-1} = \frac{10}{7} = 1.4286$

Standard Deviation
 $s_d = \sqrt{1.4286} = 1.195$

Standard Error: $SE = \frac{s_d}{\sqrt{n}}$
 $= \frac{1.195}{\sqrt{8}}$
 $= \frac{1.195}{2.828} = 0.422$

Test statistic: $t = \frac{\bar{d}}{SE}$
 $= \frac{8}{0.422} = 18.93$
 $t = 18.93$

(2) $df = n-1 = 7$
 For: $t = 18.93$, $df = 7$
 the p-value is $p < 0.0001$

(3) $p < 0.0001 < 0.05$

Question 4: Make a Decision at $\alpha=0.05$

Decision Rule:

Reject H_0 :

If $p < 0.05$

Since $p < 0.0001$

We reject the null hypothesis.

Conclusion: There is sufficient statistical evidence at the 5% significance level to conclude that the marketing campaign significantly increased sales.

Problem 13: Call Centre Response Time

Code:

```
import math

mean = 4.8
sd = 1.5
n = 100
z = 1.96

se = sd / math.sqrt(n)
me = z * se

lower = mean - me
upper = mean + me

print("Estimated Population Mean =", mean)
print("Standard Error =", se)
print("95% Confidence Interval = (", lower, ", ", upper, ")")
```

```
PS C:\Users\lenovo\OneDrive\Documents\FODS> c::; cd 'c:\Users\lenovo\OneDrive\Documents\FODS'; & 'c:\Users\lenovo\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '58547' '--' 'c:\Users\lenovo\OneDrive\Documents\FODS\C03-AT2-13.py'
Estimated Population Mean = 4.8
Standard Error = 0.15
95% Confidence Interval = ( 4.506 , 5.093999999999999 )
```

Given Data: Sample size: $n=100$

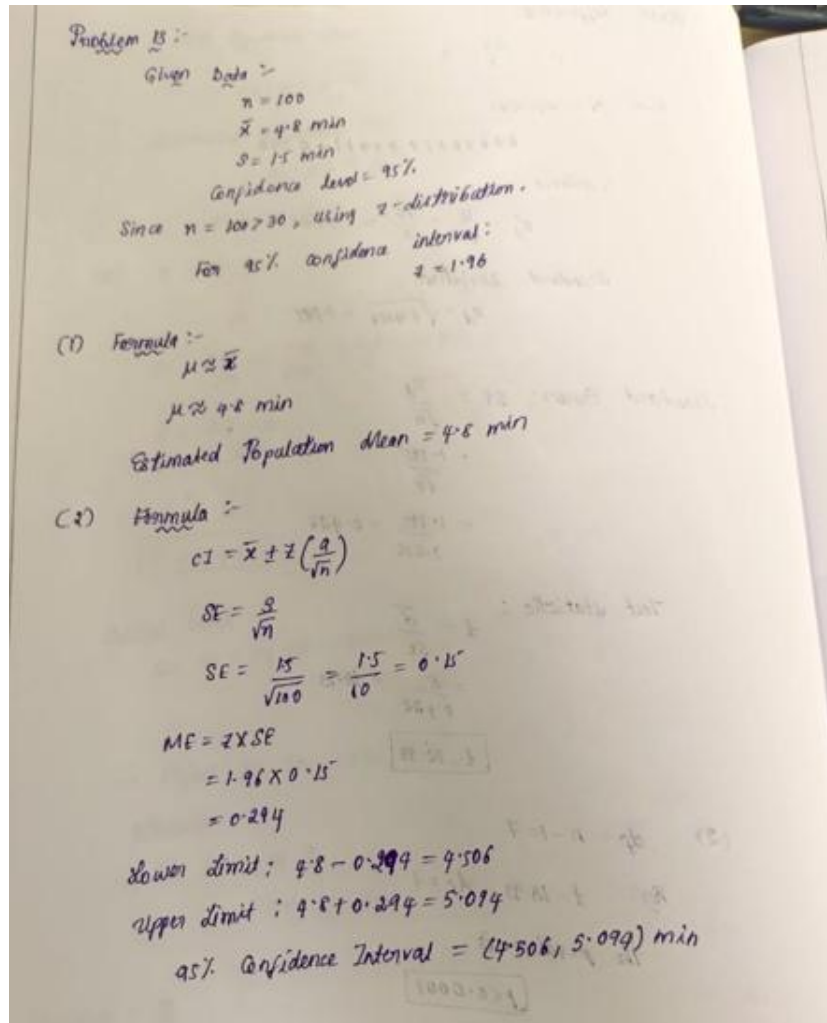
Sample mean: $\bar{X} = 4.8$ minutes

Sample standard deviation: $s=1.5$ minutes

Confidence Level = 95%

Since $n=100>30$, we use the Z-distribution.

For a 95% confidence interval: $Z=1.96$



Question 3: Explain the Frequentist Interpretation

The 95% confidence interval does not mean there is a 95% probability that the true population mean lies within 4.51 to 5.09 minutes.

Instead, it means: If many random samples of 100 calls were taken and a 95% confidence interval was calculated from each sample, approximately 95% of those intervals would contain

the true population mean response time. Thus, we are 95% confident that the true average response time lies between 4.51 and 5.09 minutes.

Problem 14: Defective Products

Code:

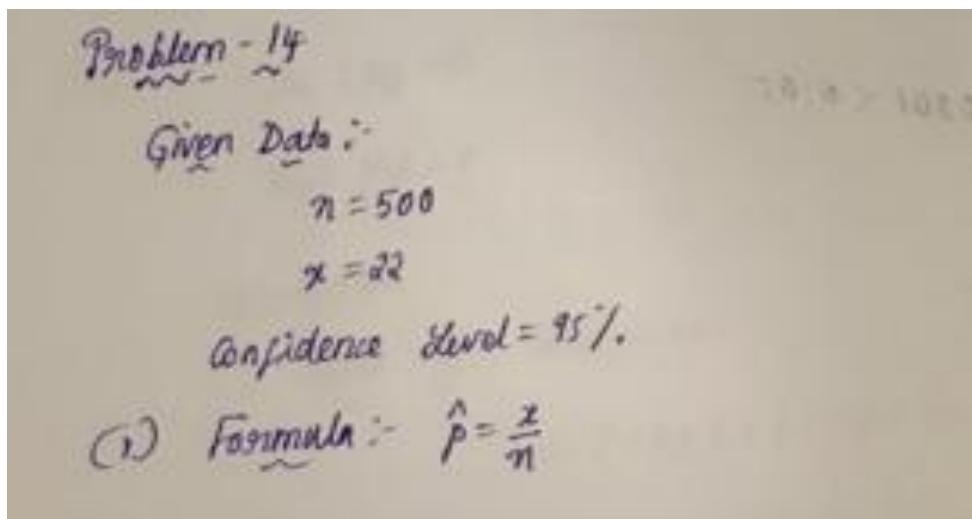
```
import math

n = 500
d = 22
z = 1.96

p = d / n
se = math.sqrt((p * (1 - p)) / n)
me = z * se

lower = p - me
upper = p + me

print("Defect Rate =", p)
print("95% Confidence Interval = (", lower, ",", upper, ")")
PS C:\Users\lenovo\OneDrive\Documents\FODS> c:: cd 'c:\Users\lenovo\OneDrive\Documents\FODS'; & 'C:\Users\lenovo\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '58600' '--' 'c:\Users\lenovo\OneDrive\Documents\FODS\CO3-AT2-14.py'
Defect Rate = 0.044
95% Confidence Interval = ( 0.02602262185968154 , 0.06197737814031845 )
```



$$\begin{aligned}
 \hat{p} &= \frac{22}{500} \\
 &= 0.044 \\
 &= 0.044 \times 100 = 4.4\% \\
 \boxed{\hat{p} = 4.4\%} \\
 (2) \quad SE &= \sqrt{\frac{0.044(1-0.044)}{500}} \\
 &= \sqrt{\frac{0.044 \times 0.956}{500}} \\
 &= \sqrt{\frac{0.042064}{500}} \\
 &= \sqrt{0.000084128} \\
 &= 0.00917 \\
 ME &= 1.96 \times 0.00917 \\
 &= 0.01797 \\
 \text{Lower Limit} &= 0.044 - 0.01797 = 0.02603 \\
 \text{Upper Limit} &= 0.044 + 0.01797 = 0.06197 \\
 &2.60\% \text{ to } 6.20\% \\
 95\% \text{ Confidence Interval} &= (0.026, 0.062)
 \end{aligned}$$

Question 3: Explain the Frequentist Meaning of the Interval

The 95% confidence interval means that if many random samples of 500 products were selected and a confidence interval was computed from each sample, then about 95% of those intervals would contain the true population defect rate.

It does not mean there is a 95% probability that the true defect rate lies between 2.6% and 6.2%. Instead, the confidence refers to the long-run performance of the confidence interval procedure.

Problem 15: Email Spam Detection

Code:

```

spam = 0.25
offer_spam = 0.80
offer_notspam = 0.10

not_spam = 1 - spam

p_offer = (offer_spam * spam) + (offer_notspam * not_spam)
p_spam_offer = (offer_spam * spam) / p_offer

print("P(Spam | Offer) =", p_spam_offer)

```

```
PS C:\Users\lenovo\OneDrive\Documents\FODS> c:: cd "C:\Users\lenovo\OneDrive\Documents\FODS" ; & "C:\Users\lenovo\AppData\Local\Programs\Python\Python313\python.exe" "c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher" "50817" "--" "c:\Users\lenovo\OneDrive\Documents\FODS\C03-AT2-15.py"
P(Spam | Offer) = 0.7272727272727273
PS C:\Users\lenovo\OneDrive\Documents\FODS>
```

Problem 15 :-

$$P(S) = 0.25$$

$$P(O|S) = 0.80$$

$$P(O|S') = 0.10$$

(i) Bayes Theorem :-

$$P(S|O) = \frac{P(O|S) \times P(S)}{P(O)}$$

Where

$$P(O) = P(O|S)P(S) + P(O|S')P(S')$$

$$P(S') = 1 - P(S)$$

$$= 1 - 0.25$$

$$= 0.75$$

$$P(O) = 0.80(0.25) + 0.10(0.75)$$

$$= 0.20 + 0.075$$

$$= 0.275$$

$$P(S|O) = \frac{0.80 \times 0.25}{0.275}$$

$$= \frac{0.20}{0.275}$$

$$= 0.7273$$

$$P(\text{Spam} | \text{Offer}) = 0.7273 = 72.73\%$$

Question 3: Interpret the Result

The probability that an email is spam, given that it contains the keyword "Offer", is approximately: 72.73%

This means: If an email contains the word "Offer", there is about a 72.73% chance that it is spam.

The keyword "Offer" is therefore a strong indicator of spam, although it does not guarantee that every such email is spam.

Problem 16: Disease Diagnosis

Code:

```
disease = 0.02
sensitivity = 0.95
specificity = 0.90

no_disease = 1 - disease
false_positive = 1 - specificity

positive = (sensitivity * disease) + (false_positive * no_disease)

posterior = (sensitivity * disease) / positive

print("P(Disease | Positive Test) =", posterior)
```

PS C:\Users\lenovo\OneDrive\Documents\FODS> c:: cd 'c:\Users\lenovo\OneDrive\Documents\FODS'; & 'C:\Users\lenovo\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '50165' '--' 'c:\Users\lenovo\OneDrive\Documents\FODS\C03-AT2-16.py'

P(Disease | Positive Test) = 0.16239316239316243

PS C:\Users\lenovo\OneDrive\Documents\FODS> █

Problem 16 :-

Disease prevalence : $P(D) = 2\% = 0.02$

Test Sensitivity : $P(+|D) = 95\% = 0.95$

Test Specificity : $P(-|D') = 90\% = 0.90$

(i) to find : $P(D|+)$

$$P(D|+) = \frac{P(+|D) \times P(D)}{P(+)}$$

$$P(+) = P(+|D)P(D) + P(+|D')P(D')$$

$$P(D') = 1 - P(D)$$

$$= 1 - 0.02$$

$$= 0.98$$

Specificity = 90%

$$P(+|D') = 1 - 0.90 = 0.10$$

$$P(+) = 0.95(0.02) + 0.10(0.98)$$

$$= 0.019 + 0.098$$

$$= 0.117$$

Applying Bayes' Theorem

$$P(D|+) = \frac{0.95 \times 0.02}{0.117} = \frac{0.019}{0.117} = 0.1624$$

$$P(D|+) = 0.1624 = 16.24\%$$

Question 3: Explain Why the Result Differs from 95%

Although the test sensitivity is 95%, the probability that a person actually has the disease after testing positive is only 16.24%.

This happens because: The disease is rare, with a prevalence of only 2%.

Most people tested do not have the disease. Even with 90% specificity, some healthy individuals receive false-positive results. Since healthy people greatly outnumber diseased people, these false positives reduce the probability that a positive test indicates actual disease. Therefore, the posterior probability (16.24%) is much lower than the test sensitivity (95%).

Problem 17: Website Conversion Rate

Code:

```
import math

n1 = 5000
x1 = 250

n2 = 5200
x2 = 312

p1 = x1 / n1
p2 = x2 / n2

p = (x1 + x2) / (n1 + n2)

se = math.sqrt(p * (1 - p) * ((1 / n1) + (1 / n2)))

z = (p2 - p1) / se

print("Conversion Rate A =", p1)
print("Conversion Rate B =", p2)
print("Z Value =", z)

if z > 1.645:
    print("Reject Null Hypothesis")
    print("Page B performs better.")
else:
    print("Fail to Reject Null Hypothesis")
```

```

PS C:\Users\lenovo\OneDrive\Documents\FODS> c:: cd 'c:\Users\lenovo\OneDrive\Documents\FODS'; & 'C:\Users\lenovo\AppData\
rams\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\de
cher' '58948' '--' 'c:\Users\lenovo\OneDrive\Documents\FODS\C03-AT2-17.py'
Conversion Rate A = 0.05
Conversion Rate B = 0.06
Z Value = 2.212713273582879
Reject Null Hypothesis
Page B performs better.
PS C:\Users\lenovo\OneDrive\Documents\FODS>

```

Problem 13 :-

(1) Formula :- $\hat{p} = \frac{x}{n}$ (Group A)

$$\hat{p}_A = \frac{250}{5000} = 0.05 = 5\%$$

Group B :- $\hat{p}_B = \frac{312}{5200} = 0.06 = 6\%$

A → 5%
B → 6%

(2) Null Hypothesis
 $H_0: p_A = p_B$

Alternative Hypothesis
 $H_1: p_B > p_A$

(3) Formula :- $\hat{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{250 + 312}{5000 + 5200}$

$$= \frac{562}{10200} = 0.0551$$

Formula :- $SE = \sqrt{\hat{p}(1-\hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$

$$= \sqrt{0.0551(0.9449) \left(\frac{1}{5000} + \frac{1}{5200} \right)}$$

$$= \sqrt{0.0521 \times 0.0003923}$$

$$= \sqrt{0.00002045}$$

$$= 0.00452$$

Calculate z Statistic

Formula :- $z = \frac{\hat{p}_B - \hat{p}_A}{SE} = \frac{0.06 - 0.05}{0.00452} = \frac{0.01}{0.00452} = 2.21$

$z = 2.21$

Critical value :-

For $\alpha = 0.05$ (one-tailed test)

$z_{critical} = 1.645$

$2.21 > 1.645$

Question 4: Determine Whether Page B Performs Better

Since $Z=2.21 > 1.645$ we reject the null hypothesis.

Conclusion

There is sufficient statistical evidence at the 5% significance level to conclude that Landing Page B has a significantly higher conversion rate than Landing Page A.

Problem 18: Mobile App Feature Test

Code:

```
import math

n1 = 400
m1 = 18
sd1 = 5

n2 = 420
m2 = 20
sd2 = 6

se = math.sqrt((sd1**2/n1) + (sd2**2/n2))
z = (m2 - m1) / se

print("Z Value =", z)

if z > 1.645:
    print("Reject Null Hypothesis")
    print("Feature B increases engagement.")
else:
    print("Fail to Reject Null Hypothesis")

PS C:\Users\lenovo\OneDrive\Documents\FODS> c;; cd 'c:\Users\lenovo\OneDrive\Documents\FODS'; & 'C:\Users\lenovo\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '57054' '--' 'c:\Users\lenovo\OneDrive\Documents\FODS\CO3-AT2-18.py'
Z Value = 5.194992955455177
Reject Null Hypothesis
Feature B increases engagement.
PS C:\Users\lenovo\OneDrive\Documents\FODS> 
```

Problem 18 :-

Feature A :-

$n_1 = 400$

$\bar{x}_1 = 18 \text{ min}$

$s_1 = 5 \text{ min}$

Feature B :-

$n_2 = 420$

$\bar{x}_2 = 20 \text{ min}$

$s_2 = 6 \text{ min}$

$\alpha = 0.05$

(1) Null Hypothesis :-

$$H_0 = \mu_B \leq \mu_A$$

Alternative Hypothesis :-

$$H_1 = \mu_B > \mu_A$$

(2)
$$Z = \frac{\bar{x}_2 - \bar{x}_1}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$SE = \sqrt{\frac{5^2}{400} + \frac{6^2}{420}} = \sqrt{\frac{25}{400} + \frac{36}{420}} = \sqrt{0.0625 + 0.0857}$$

$$= \sqrt{0.1482} = 0.385$$

$\text{Diff} = 20 - 18 = 2$

$$Z = \frac{2}{0.385} = 5.19$$

For $Z = 5.19$

$p < 0.00001$

Conclusion

There is strong statistical evidence that Feature B significantly increases user engagement compared to Feature A.

Feature B should therefore be preferred for improving user engagement.

Problem 19: Recommendation System

Code:

```
import math

n1 = 100000
n2 = 100000

p1 = 0.048
p2 = 0.053

x1 = n1 * p1
x2 = n2 * p2

p = (x1 + x2) / (n1 + n2)

se = math.sqrt(p * (1 - p) * ((1/n1) + (1/n2)))
z = (p2 - p1) / se

print("Z Value =", z)

if z > 1.645:
    print("Reject Null Hypothesis")
    print("New Recommendation System is better.")
else:
    print("Fail to Reject Null Hypothesis")
```

```
PS C:\Users\lenovo\OneDrive\Documents\FODS> c:: cd 'c:\Users\lenovo\OneDrive\Documents\FODS'; & 'C:\Users\lenovo\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher' '65182' '--' 'c:\Users\lenovo\OneDrive\Documents\FODS\CO3-AT2-19.py'
Z Value = 5.105776885711038
Reject Null Hypothesis
New Recommendation System is better.
PS C:\Users\lenovo\OneDrive\Documents\FODS> 
```

Problem 19:

(1) $H_0: p_1 = p_2$
 $H_1: p_2 > p_1$

$p_1 = 4.8\% = 0.048$
 $p_2 = 5.8\% = 0.058$

$x_1 = 100000 \times 0.048 = 4800$
 $x_2 = 100000 \times 0.058 = 5800$

$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$
 $= \frac{4800 + 5800}{100000 + 100000}$
 $= \frac{10600}{200000} = 0.053$

$SE = \sqrt{\hat{p}(1-\hat{p})\left[\frac{1}{n_1} + \frac{1}{n_2}\right]}$
 $SE = \sqrt{0.053(0.947)\left(\frac{1}{100000} + \frac{1}{100000}\right)}$
 $= \sqrt{0.04795 \times 0.00002}$
 $= \sqrt{0.000000959}$
 $= 0.000979$

$Z = \frac{p_2 - p_1}{SE} = \frac{0.058 - 0.048}{0.000979} = \frac{0.01}{0.000979} = 10.21$
 $Z = 5.11$

Question 3: Discuss Practical Significance vs Statistical Significance

Statistical Significance

Since $Z=5.11$

$p < 0.00001$ the difference between the two recommendation systems is statistically significant.

This means the increase in click-through rate is very unlikely to have occurred by chance.

Problem 20: Salary Prediction Model

Code:

```
import math

mean = 200
sd = 350
n = 50

se = sd / math.sqrt(n)
t = mean / se

print("t Value =", t)

if t > 1.676:
    print("Reject Null Hypothesis")
    print("Model B performs better.")
else:
    print("Fail to Reject Null Hypothesis")
```

```
PS C:\Users\lenovo\OneDrive\Documents\FODS> c:; cd 'c:\Users\lenovo\OneDrive\Documents\FODS'; & 'c:\Users\lenovo\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher' '65249' '--' 'c:\Users\lenovo\OneDrive\Documents\FODS\CO3-AT2-20.py'
t Value = 4.040610178208843
Reject Null Hypothesis
Model B performs better.
PS C:\Users\lenovo\OneDrive\Documents\FODS> 
```

Problem 20

Model A MAE = 4200
 Model B MAE = 4000

$\bar{d} = 200$
 $s_d = 350$
 $n = 50$

(1) $H_0: \mu_d = 0$
 $H_A: \mu_d > 0$

$$t = \frac{\bar{d}}{\frac{s_d}{\sqrt{n}}}$$

$$SE = \frac{s_d}{\sqrt{n}}$$

$$SE = \frac{350}{\sqrt{50}}$$

$$= \frac{350}{7.071}$$

$$= 49.50$$

$$t = \frac{200}{49.50} = 4.04$$

$t = 4.04$

$$df = n - 1 = 49$$

Question 3: Explain the Result in Terms of Statistical Significance

Since $t=4.04$

$p < 0.001$ there is strong statistical evidence that Model B performs better than Model A.

Model B has a lower Mean Absolute Error (4000) compared to Model A (4200), indicating that its predictions are more accurate.

Therefore, the observed improvement is unlikely to be due to random variation.